

International Finance

Lecture IV– March 10th

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

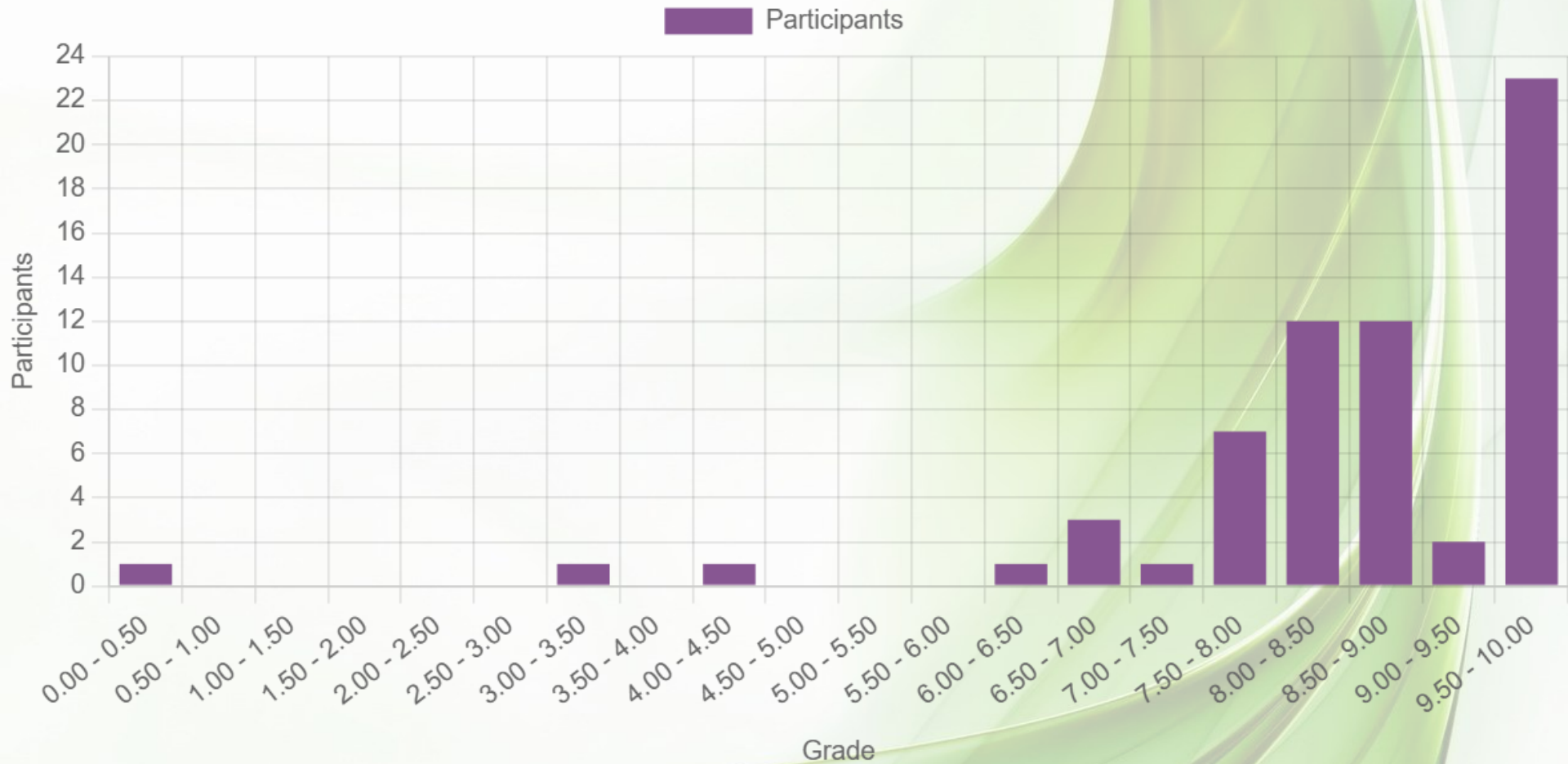
(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

Outline

- Course organization
 - Attendance, quiz
 - Quiz #4: will be open by tomorrow (Wednesday) midnight
 - The usual deadline: next Monday midnight
 - Quiz #3: closed a bit early
- Asset approach to exchange rates
 - Uncovered interest rate parity (UIP)
- Additional components:
 - Money
 - Money demand & supply
- Merging money market and UIP:
 - Short run
 - Long run
 - Overshooting

Quiz #3



Average: 8.47

73 attempts/72 participants

Reading for the Next Week:

- This lecture is largely based on Krugman & Obstfeld & Melitz: International Economics (chapters 14 & 15)
- Economist (2025) - Don't let Donald Trump see our Big Mac index
- Economist (2024): What Donald Trump can learn from the Big Mac index

Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. International transfers/payments. Channels, related instruments. (February 24th)
3. Balance of payments structure and national income accounting for an open economy. International investment position (March 3rd)
4. **Introduction to exchange rate determination: asset approach. Covered and uncovered interest rate parity.** (March 10th)
5. Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity. Overshooting. (March 17th)
6. Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect. Application on the nominal and real convergence in the EU (March 24th)
7. Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies. (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications. Interventions and sterilisation. Policy trilemma. (April 14th)
10. **Monetary integration: costs and benefits. Monetary integration: effects – empirical research.** (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. **Reserved for a make-up class (possibly for April 21st). (May 19th)**

Bellocchi & Travaglini (2025): The Trade Deficit Delusion: Why Tariffs Will Not Make America Great Again

1. How did the “war on terror” under G.W. Bush influence the trade deficit?
2. To what extent can the US trade deficits be attributed to “predatory factors” (i.e. other countries’ unfair practices)?
3. Why do authors claim that trade policies are not only ineffective for correcting trade deficits, but they can even make the deficits worse?
4. What is the “impossible theorem” mentioned in the text?
5. Which of the three scenarios outlined in the text appears to be happening?

Exchange Rates



		IT SELL	CHQ BUY	NOTE BUY
 USD	0.8142	0.8146	0.8886	0.8894
 GBP	0.5117	0.5121	0.5611	0.5674
 EUR	0.6527	0.6531	0.7225	0.7291
 NZD	1.0623	1.0634	1.1498	1.1704
 JPY	93.1470	93.4230	103.700	106.560
 HKD	6.3284	6.3330	6.8680	6.9416
 SGD	1.0456	1.0567	1.1558	1.2196

What are Exchange Rates?

- **Exchange rates** = prices of money (expressed in units of other money)
 - Do you know some other “price of money”?
- Exchange rates are quoted as
 - Domestic currency per unit of foreign currency (direct or price notation)
 - E.g. 21 CZK per 1 USD
 - Foreign currency per unit of domestic currency (indirect or volume notation)
 - E.g. 0.048 USD per 1 CZK
 - Often used in Britain
- **Exchange rate is necessarily also an asset price**
 - It is governed by similar principles as other asset prices
 - Most importantly: the value of an asset today is related to the purchasing power over goods and services in future.

“Types” of Exchange Rates

- Spot v. forward exchange rates
 - Are we pricing the foreign currency now for current (spot) transactions, or have we today agreed on an exchange rate for some transaction which will take place in future
 - More details on forwards will be provided later
- Nominal vs. real exchange rates
 - Nominal rates: relative prices of currencies
 - **Real exchange rates:** indicators based on data on nominal exchange rates and inflation rates of the respective countries
 - **Relative price of commodities/services!**
 - Price competitiveness
- Effective exchange rates
 - Weighted averages of index of individual exchange rates

$$R = S \frac{P^*}{P} = \frac{SP^*}{P}$$

Exchange Rates: Constraints

- Assume that we are analyzing exchange rates of three currencies which are traded freely, e.g. EUR, USD, CZK
- What would happen if you notice the following situation in the market:
 - CZK/EUR: 24
 - CZK/USD: 20
 - USD/EUR: 1.1

Assume that we have **22,000 CZK**...

Buy 1100 USD, trade for 1000 EUR,
and finally use the Euros to buy **24,000 CZK**

“Rinse & Repeat”

Exchange Rates: Triangular Arbitrage

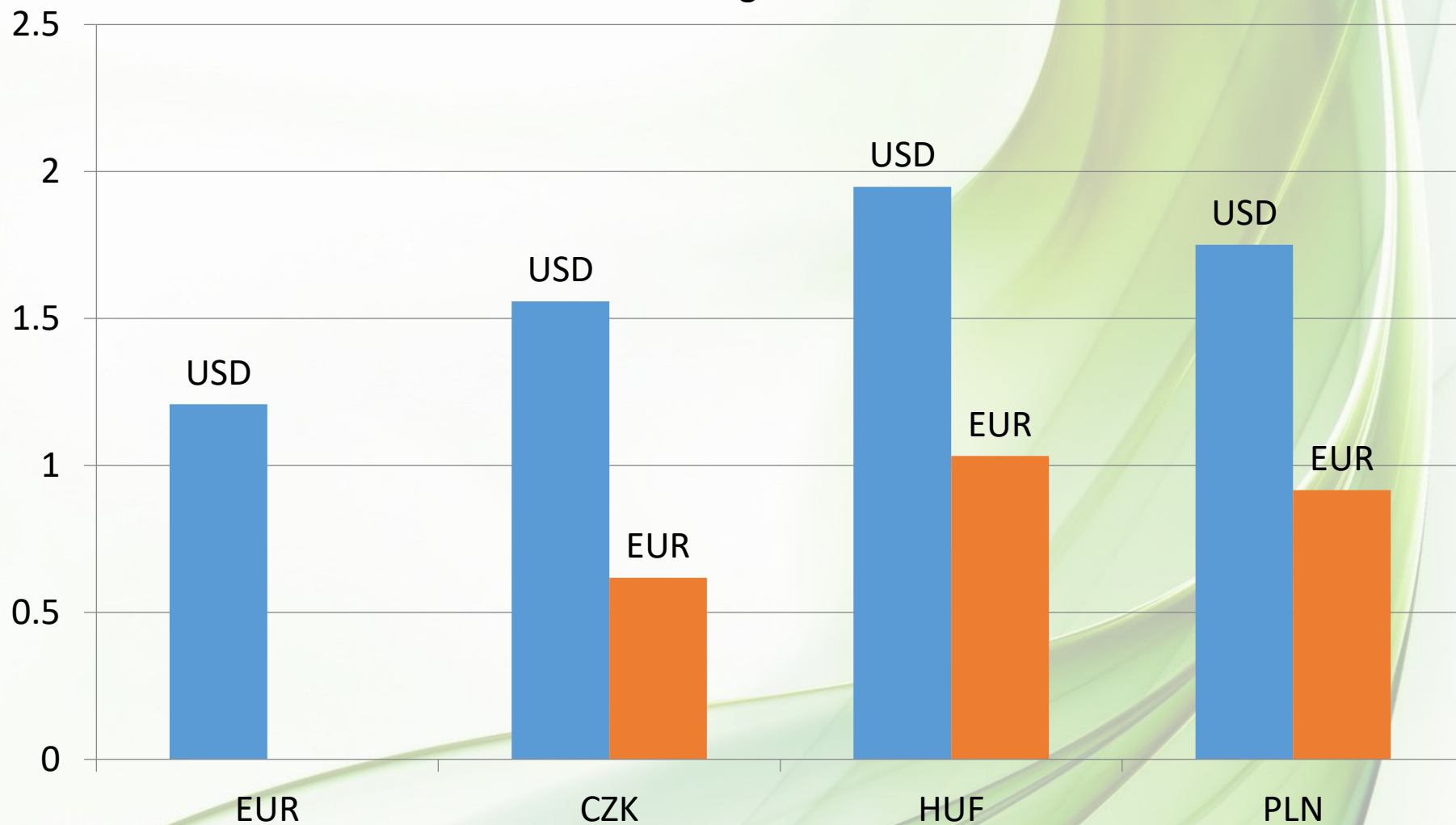
- If currencies are freely traded, if assets can be freely bought sold in such currencies and if markets are reasonably efficient...
- Then different exchange rates are not fully independent:

$$S_{CZK/USD} = S_{CZK/EUR} \cdot S_{EUR/USD}$$

- Dollar is the dominant **vehicle currency** (followed by Euro).
 - Most transactions from CZK to exotic currencies will thus be done indirectly:
 - For example: CZK → USD → BRL
 - The arbitrage conditions tie the exchange rates together

Currency Volatility, 2011, %

Average volatility based on 20-day variation of coefficients calculated from daily exchange rates.



Own calculations based on Pacific Exchange Rate data



Introduction to Asset Approach and Un/Covered Interest Parity and Forward Exchange

Exchange Rates and Interest Rates

- Two types of seemingly very similar relationships:
 - **Uncovered interest rate parity (UIP)**
 - Equilibrium relationship between spot exchange rate, expected exchange rate and interest rates
 - *Covered interest rate parity (CIP)*
 - *Arbitrage relationship between spot, forward exchange rate and interest rates*

UIP (Uncovered IR Parity)

- Uncovered interest rate parity
- When doing investment decisions, people compare return on assets in different countries (+ don't forget **risk** and **liquidity**!)
- Return on foreign asset comprises its return in foreign currency and change in exchange rate
- Simplified version:

$$\Delta E_{CZK/\text{€}}^e \text{ (rel. change)} \cong R_{CZK} - R_{\text{€}}$$

Relative change – in % if interest rates (R) are in percent.

UIP – Mathematical Details

$$Deposit \cdot (1 + R_{CZK}) = Deposit \cdot \frac{1}{E_{CZK/\text{€}}} \cdot (1 + R_{\text{€}}) \cdot E_{CZK/\text{€}}^e$$

$$\frac{(1 + R_{CZK})}{(1 + R_{\text{€}})} = \frac{E_{CZK/\text{€}}^e}{E_{CZK/\text{€}}} \quad \text{Subtract 1 from both sides...}$$

$$\frac{1 + R_{CZK} - 1 - R_{\text{€}}}{1 + R_{\text{€}}} = \frac{E_{CZK/\text{€}}^e - E_{CZK/\text{€}}}{E_{CZK/\text{€}}}$$

$$\frac{R_{CZK} - R_{\text{€}}}{1 + R_{\text{€}}} = \frac{E_{CZK/\text{€}}^e - E_{CZK/\text{€}}}{E_{CZK/\text{€}}}$$

Mostly quite small!

$$R_{CZK} - R_{\text{€}} = \frac{E_{CZK/\text{€}}^e - E_{CZK/\text{€}}}{E_{CZK/\text{€}}} + \frac{E_{CZK/\text{€}}^e - E_{CZK/\text{€}}}{E_{CZK/\text{€}}} \cdot R_{\text{€}}$$

“Interest rate parity”

$$\frac{E_{CZK/\text{€}}^e - E_{CZK/\text{€}}}{E_{CZK/\text{€}}} \cong R_{CZK} - R_{\text{€}}$$

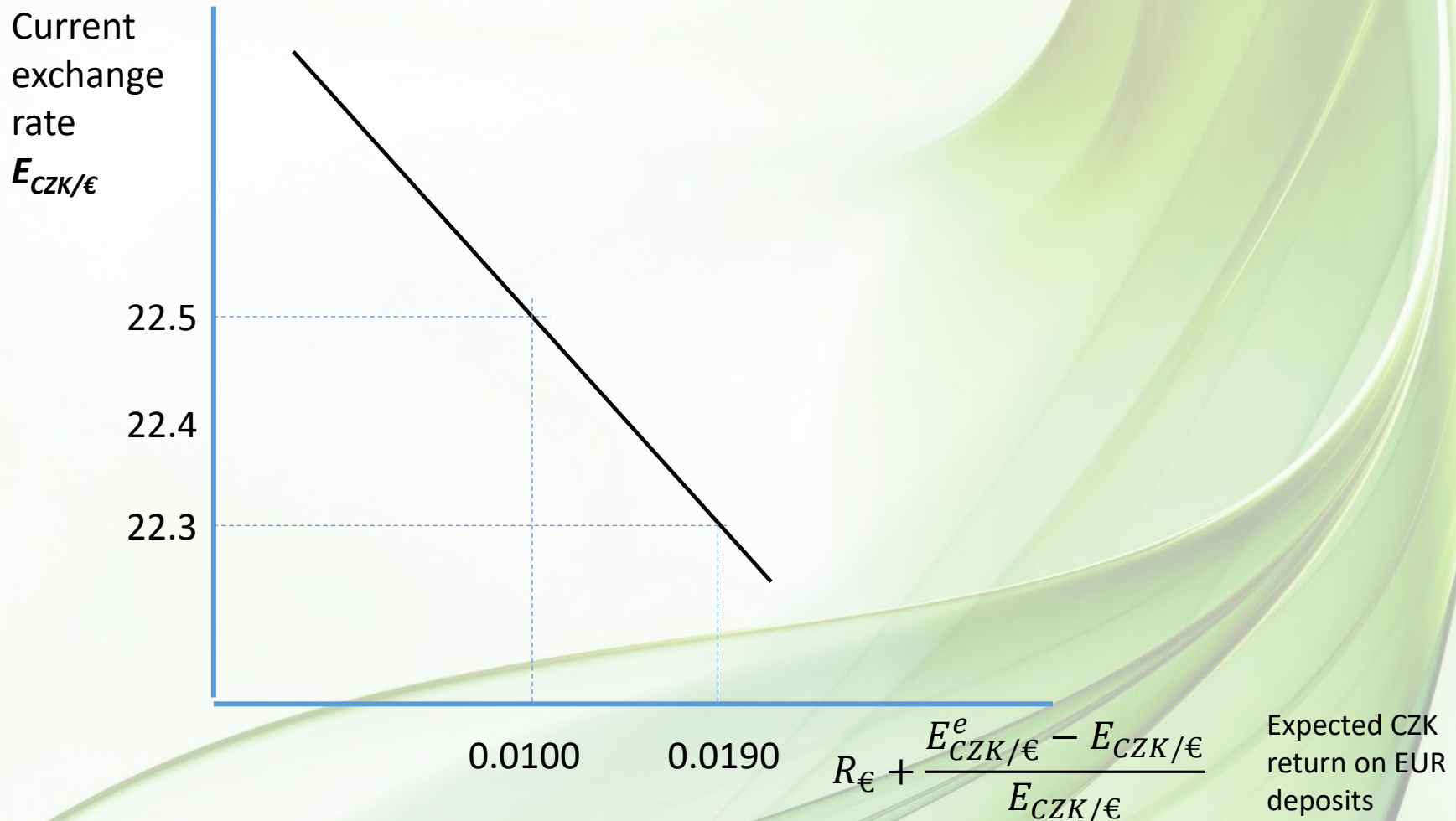
$$R_{CZK} \cong R_{\text{€}} + \frac{E_{CZK/\text{€}}^e - E_{CZK/\text{€}}}{E_{CZK/\text{€}}}$$

Expected Returns on Euro Deposits when $E^e_{CZK/€} = 22.5$ CZK Per Euro

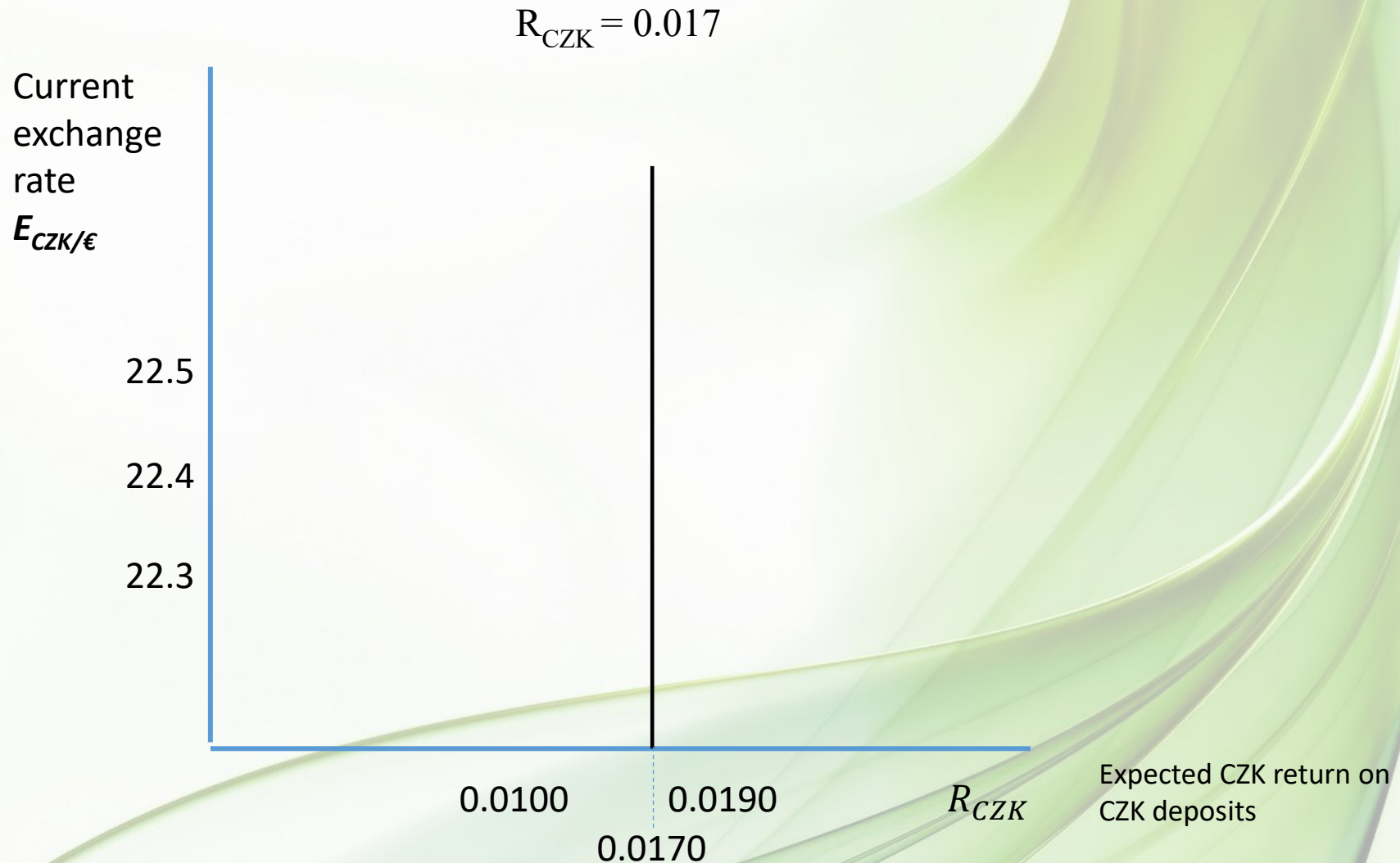
Current exchange rate	Interest rate on euro deposits	Expected rate of CZK depreciation	Expected CZK returns on euro deposits
$E_{CZK/€}$	$R_€$	$(22.5 - E_{CZK/€})/E_{CZK/€}$	$R_€ + (22.5 - E_{CZK/€})/E_{CZK/€}$
22.30	0.01	0.0090	0.0190
22.40	0.01	0.0045	0.0145
22.50	0.01	0.0000	0.0100
22.60	0.01	-0.0044	0.0056
22.70	0.01	-0.0088	0.0012

The Current Exchange Rate and the Expected Return on Euro Deposits

$$E_{CZK/\text{€}}^e = 22.5, R_{\text{€}} = 0.01$$

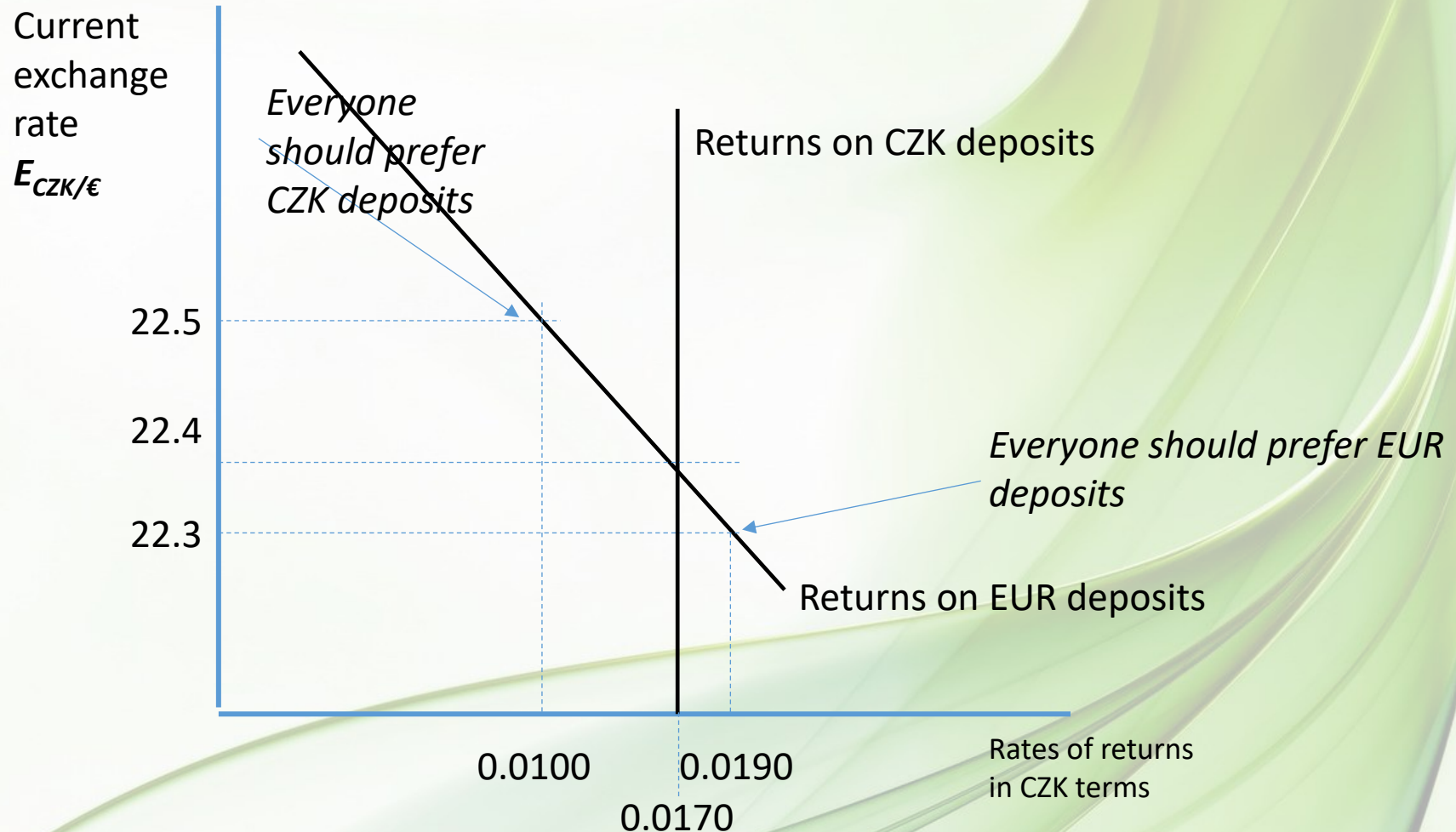


The Current Exchange Rate and the Expected Return on CZK Deposits



Determination of the “Equilibrium Exchange Rate”

$$E^e_{CZK/\epsilon} = 22.5, R_{\epsilon} = 0.01, R_{CZK} = 0.017$$



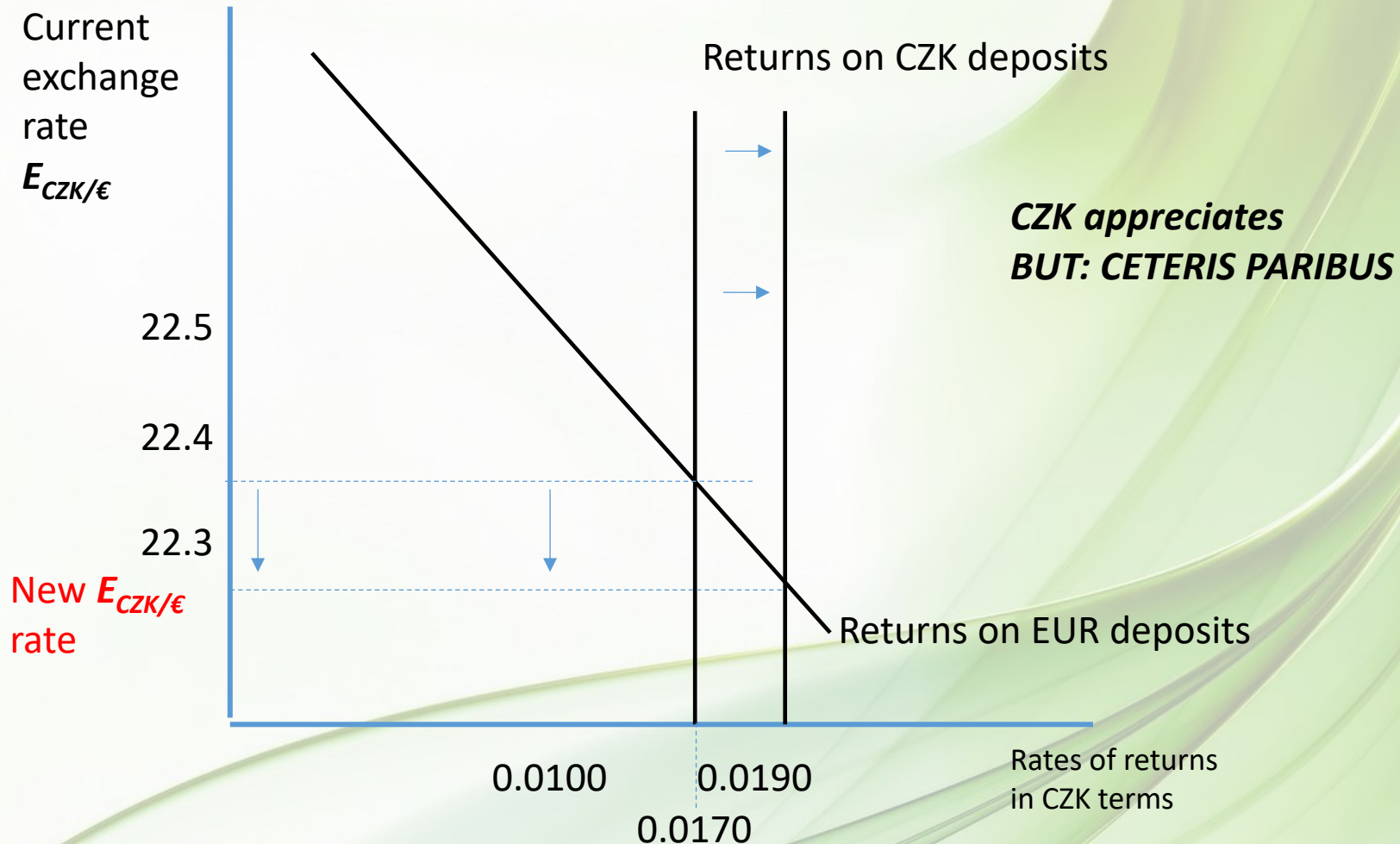
The Market for Foreign Exchange

- The effects of changing interest rates:
 - An **increase in the interest rate** paid on deposits denominated in a particular currency will increase the rate of return on those deposits. This leads to an **appreciation of the currency**.
 - A rise in CZK interest rates causes the CZK to appreciate.
 - A rise in euro interest rates causes the CZK to depreciate.

Of course: **ceteris paribus**

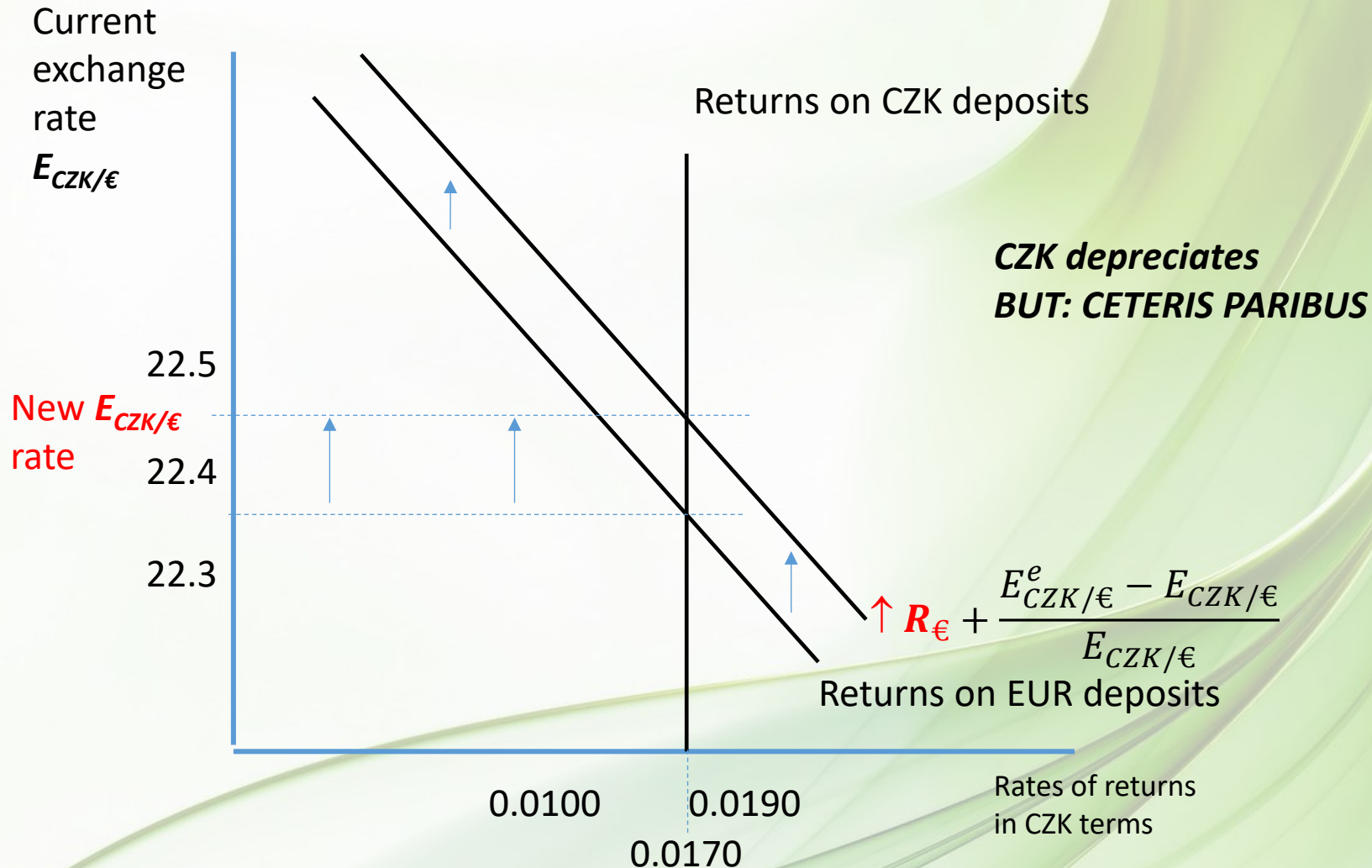
Increase in Returns on CZK Deposits

$$E_{CZK/\text{€}}^e = 22.5, R_{\text{€}} = 0.01, R_{CZK} = 0.017 \rightarrow 0.020$$



Increase in Returns on EUR Deposits

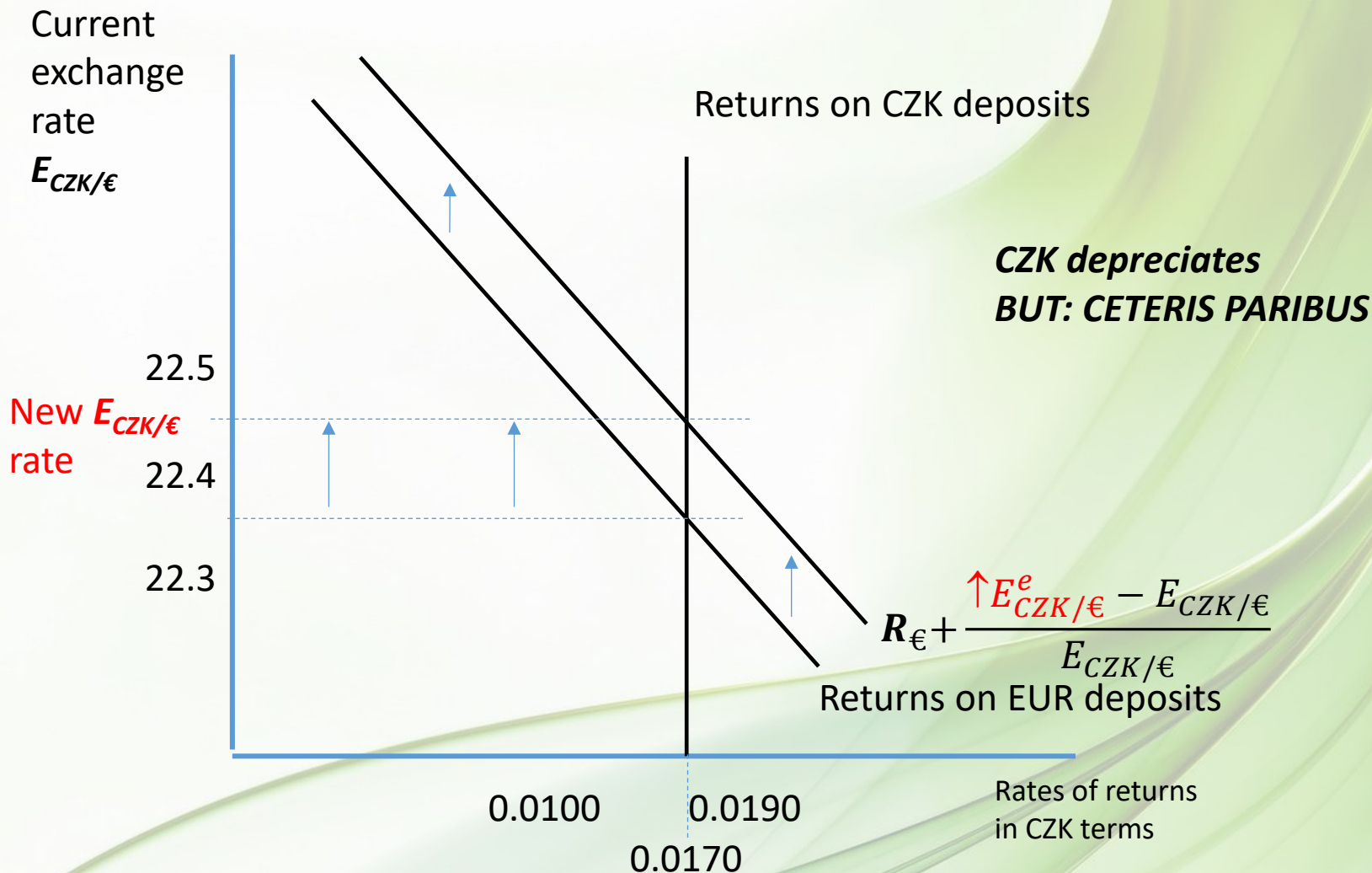
$$E^e_{CZK/\epsilon} = 22.5, R_{\epsilon} = 0.01 \rightarrow 0.015, R_{CZK} = 0.017$$



Effects of the Change in $E^e_{CZK/€}$

Expected Future Depreciation of CZK

$$E^e_{CZK/€} = 22.5 \rightarrow \mathbf{23}, R_€ = 0.01, R_{CZK} = 0.017$$



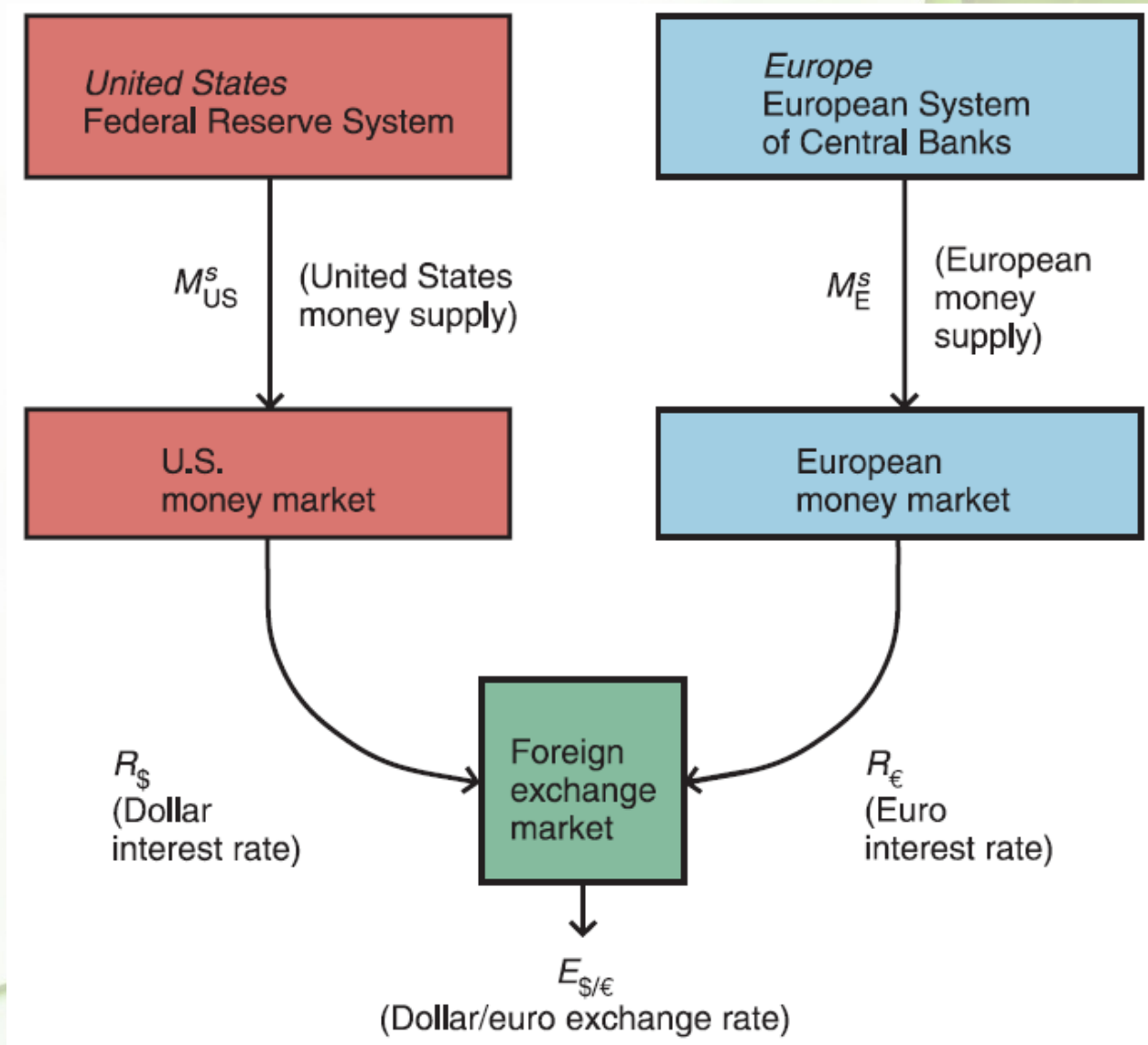
Hidden Assumptions

- There are some assumptions necessary for the simple interpretation of the charts provided in previous slides
 - **Long run exchange rate** mostly kept **stable**!
 - We assumed just one interest rate for every currency
 - No significant transaction costs or other constraints
- Also - the simple relationship would hold only:
 - If the two currencies (assets in the two currencies) have the same liquidity and the same risk

Money, Interest Rates and Exchange Rates



Money Market – Exchange Rate Linkages



Keynes and Money Demand

Keynes defined **three motives** for holding money:

1. *Transactions motive* –the use of money in regular payments
 2. *Precautionary motive* – demand for money to meet unforeseen contingencies
 3. *Speculative motive* – related to uncertainty about value of other assets
- *We will leave the actual motives to monetary economists, we will simply assume that*
 - *Money demand is the function of output (income) and interest rates*

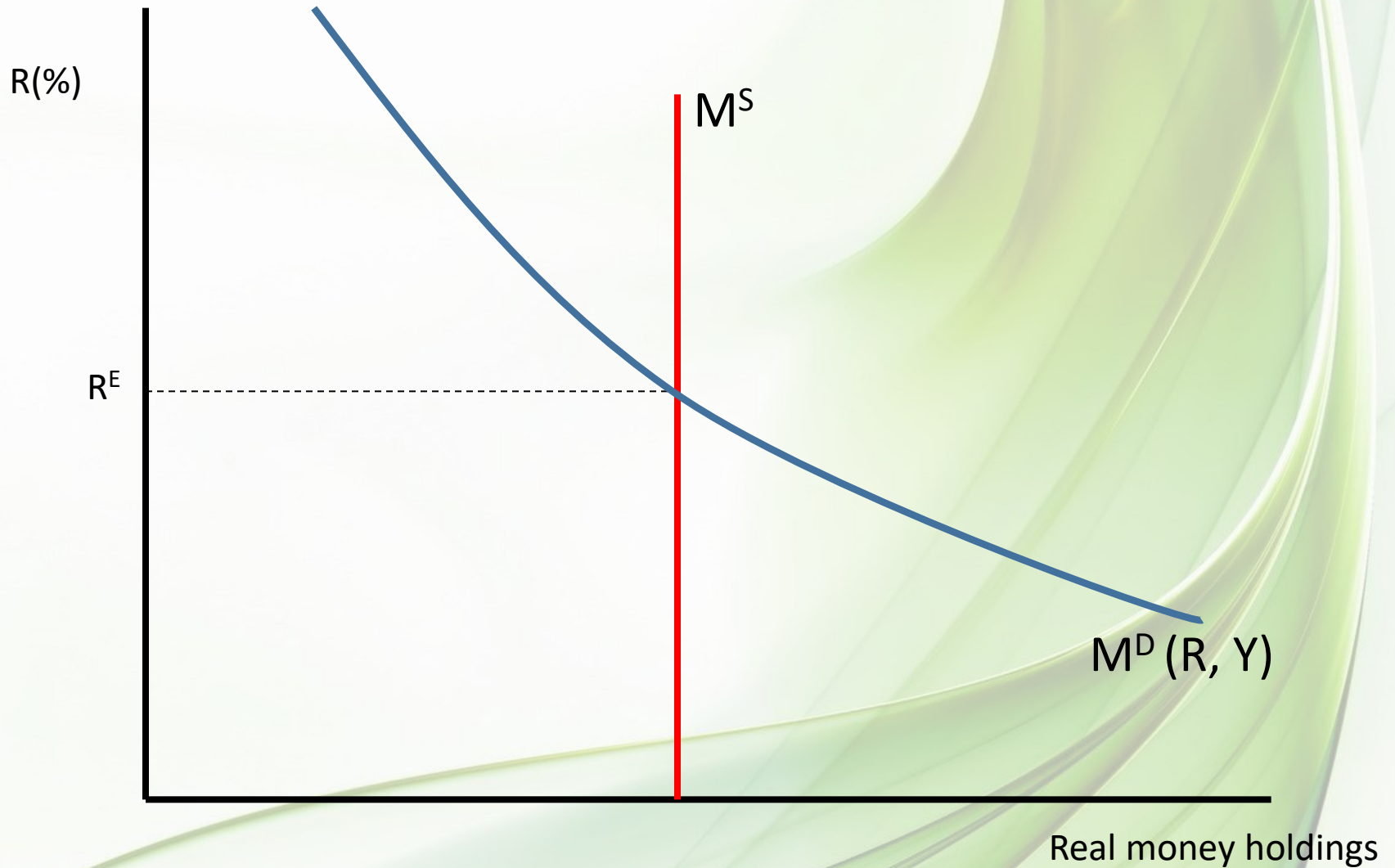
Demand for Money

- Demand for real money balances

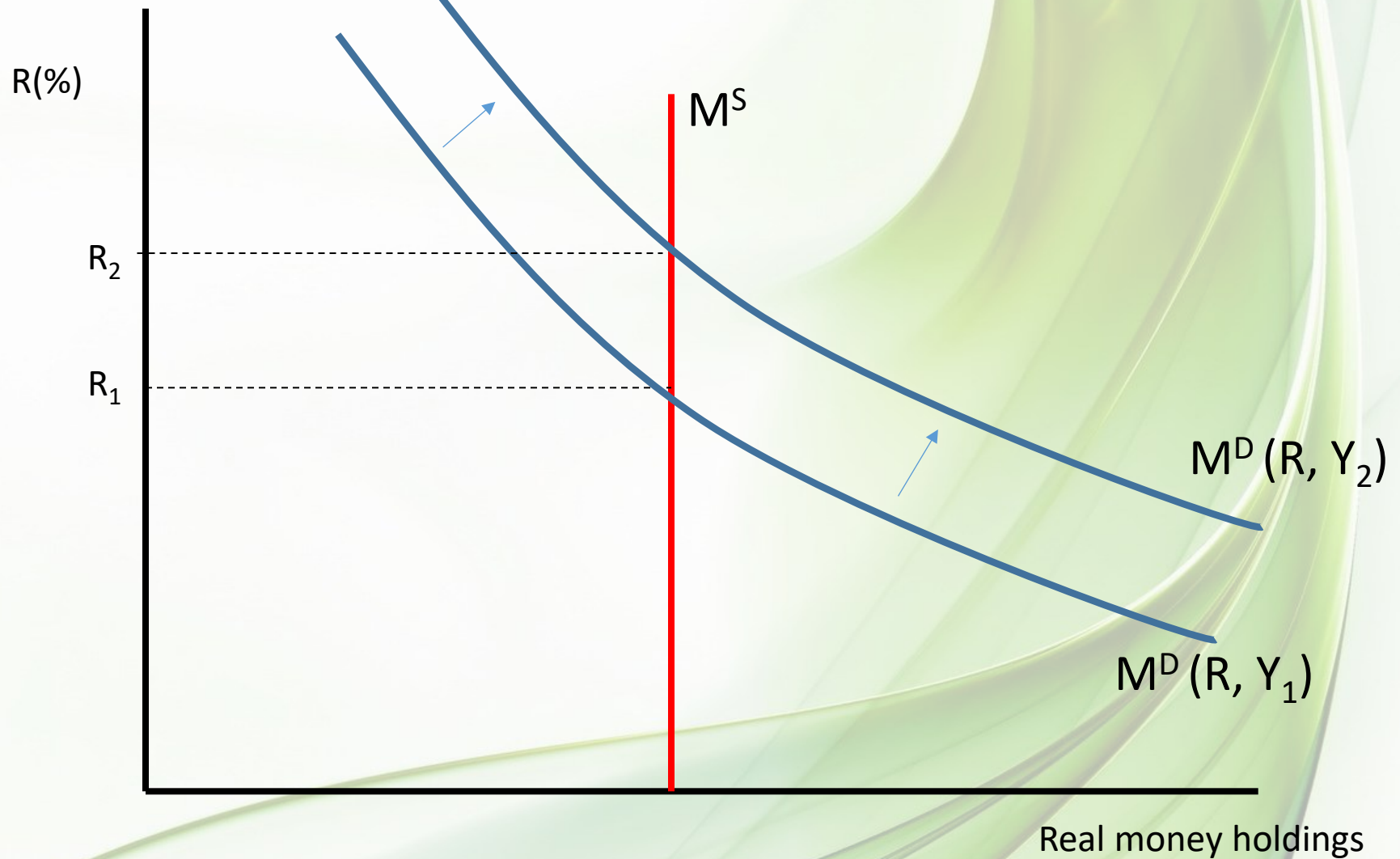
$$\frac{M^D}{P} = L(\underset{+}{Y}, \underset{-}{i})$$

- Factors:
 - Income effect (wealth effect, transaction demand)
 - Price effect – people care about real money balances (Keynes)
 - Interest rate – related to opportunity costs of holding money
 - Speculative demand?

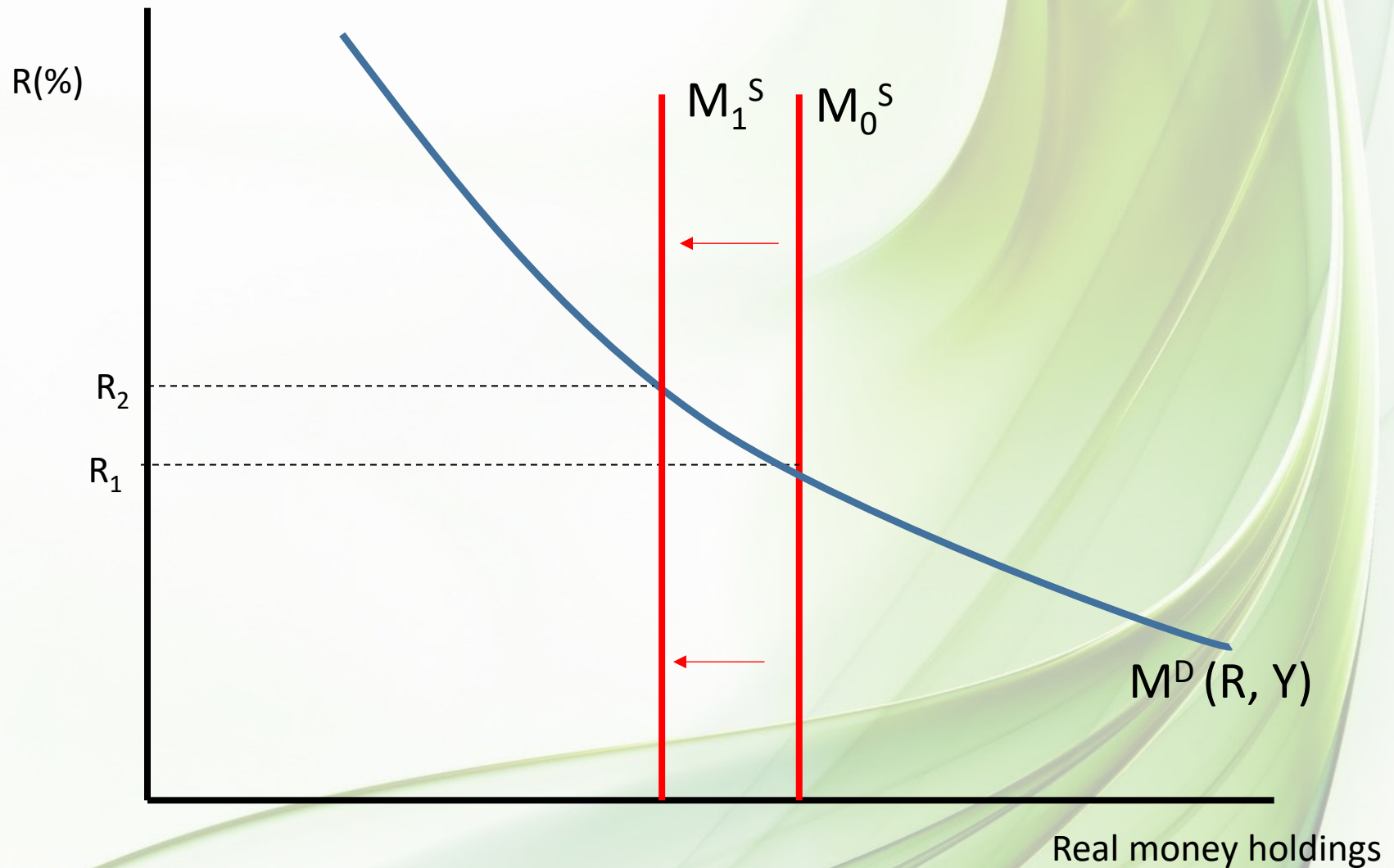
Money Market Equilibrium



Comparative Statics: Increase in Our Real Income

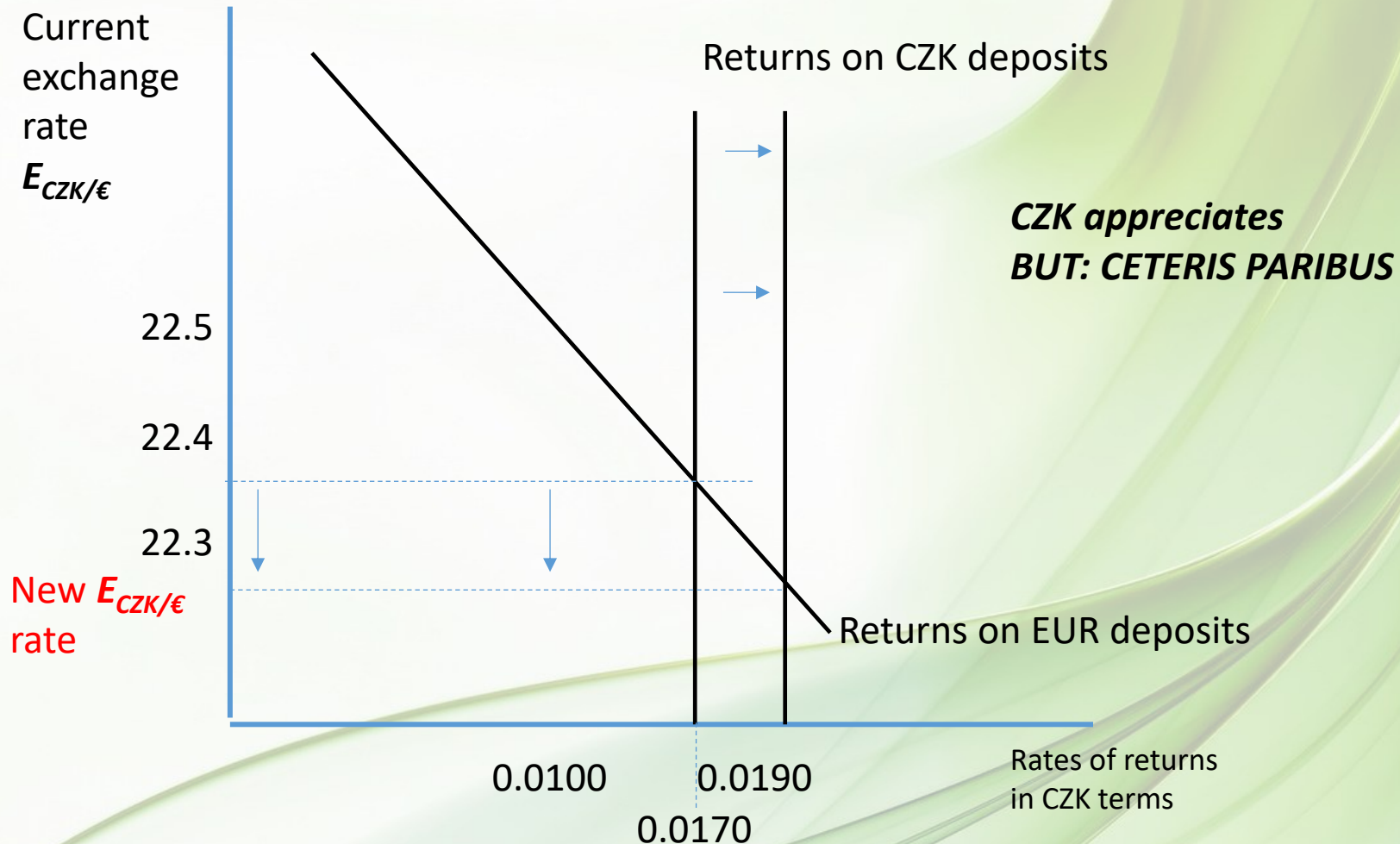


Comparative Statics: Decrease in Our Real Money Supply



Increase in Returns on CZK Deposits

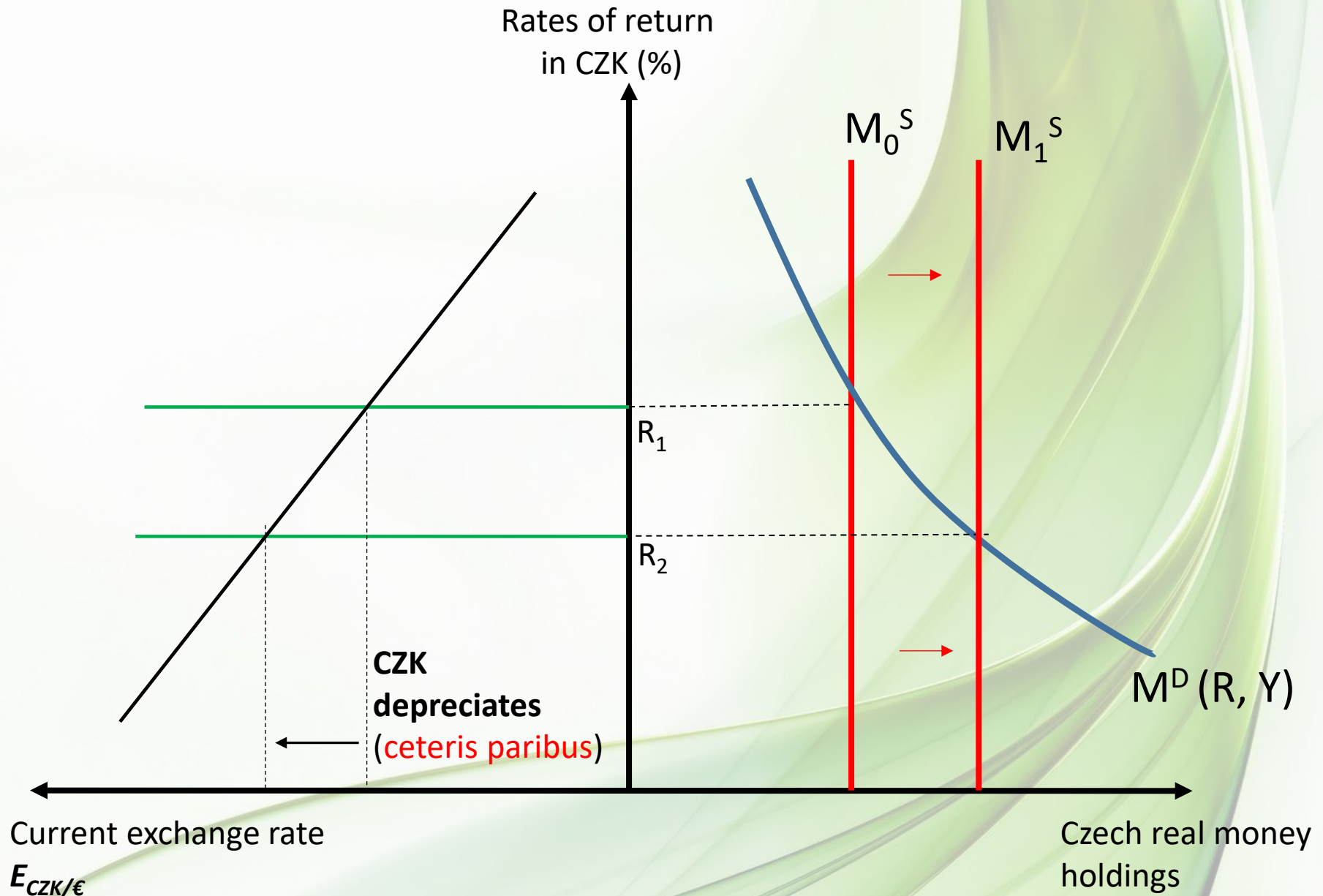
$$E^e_{\text{CZK}/\text{€}} = 22.5, R_{\text{€}} = 0.01, R_{\text{CZK}} = 0.017 \rightarrow 0.020$$



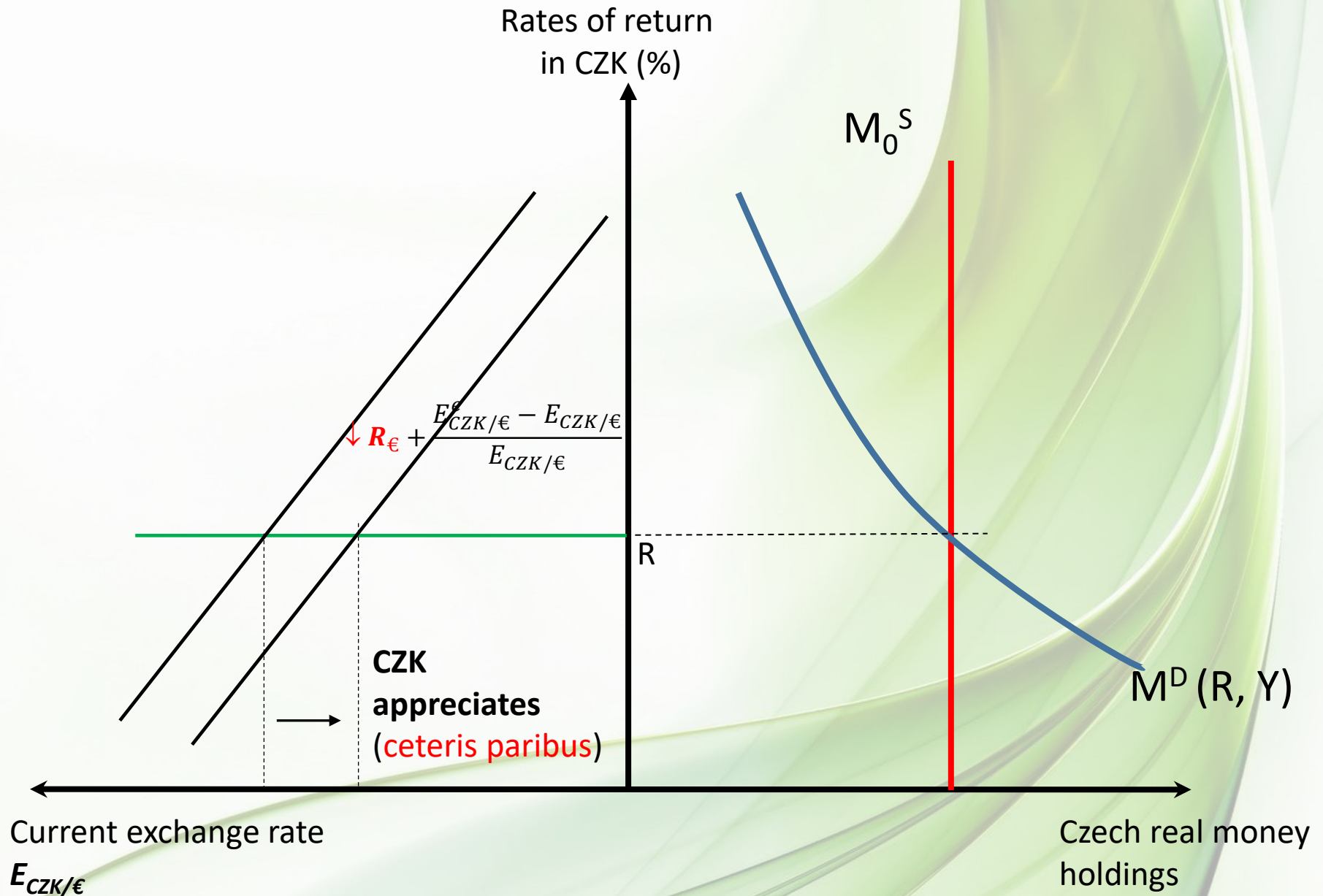
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Let us Connect the Two Types of
Diagrams...

Increase in the Our Real Money Supply



Increase in the EMU Real Money Supply



Money Supply and Exchange Rates

- Monetary policies influence not only interest rates but they have also implications for
 - **Price levels** (mainly in the long run – due to rigidities)
 - Real money balances (M/P)
 - Expected future levels of exchange rates (if the **PPP** is valid in the long run)
- We should try to reconcile the **long** and short run into some coherent frameworks

Digression (1): Purchasing Power Parity

- **Absolute PPP**

- Basic logic: **law of one price**
- The currency should have the same purchasing power at home and abroad

$$P_{CZ} = P_{GER} \cdot S_{CZK/EUR} \Rightarrow$$

$$S_{CZK/EUR} = \frac{P_{CZ}}{P_{GER}}$$

- **Logic:** if PPP does not hold, we should observe large imbalance in the balance of trade, then the exchange rate would adjust back to PPP

$$S > S_{PPP} \Rightarrow X > M \Rightarrow S \downarrow$$

$$S < S_{PPP} \Rightarrow M > X \Rightarrow S \uparrow$$

Digression (2): Quantity Theory of Money

- Original version (Fisher, 1911):

$$M \cdot V = P \cdot T$$

- Newer version:

$$M \cdot V_Y = P \cdot Y$$

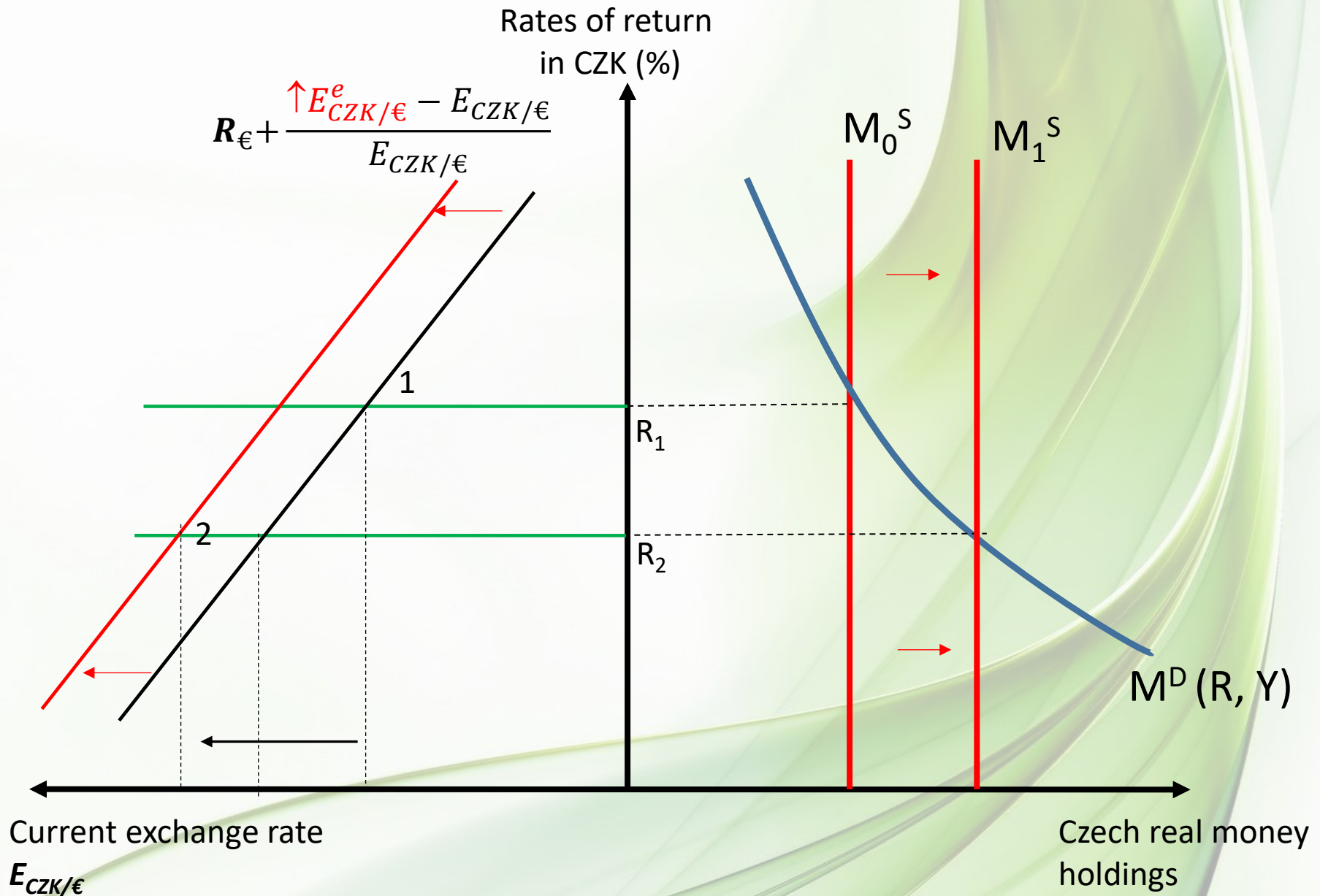
- **Traditional interpretation:**

- Growth of money supply leads to inflation, i.e. higher price level in future
- Holds always as an identity
- Some issues:
 - Is the velocity stable?

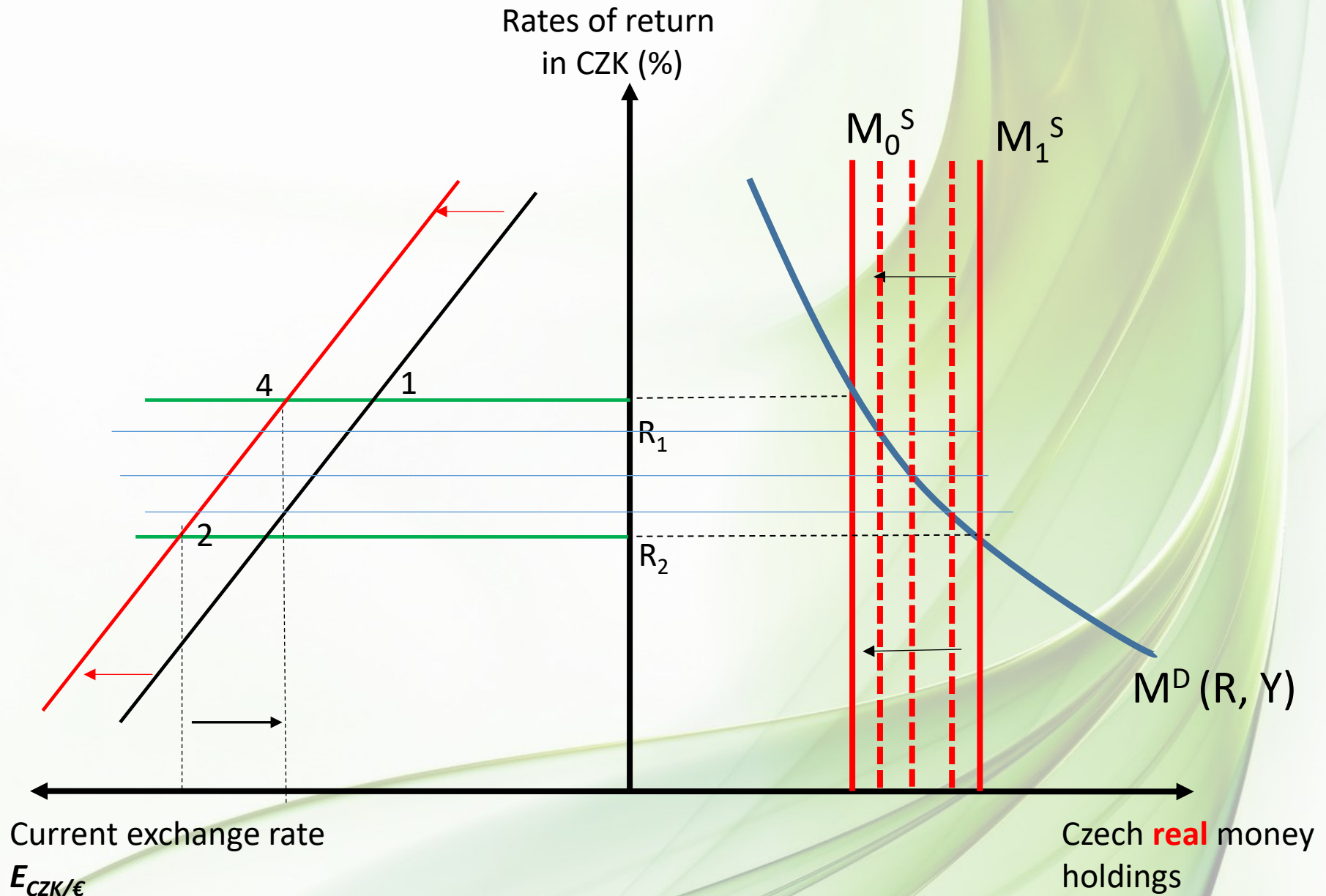
Putting the Two Digressions Together

- If Czech central bank increases domestic money supply (ceteris paribus):
 - Sooner or later domestic prices will increase
 - Higher domestic prices imply depreciation in the long run
- But our previous analysis of the effects of changes in domestic money supply on exchange rate assume that the expected future exchange rate was stable
 - So – we must redo the analysis assuming the expected exchange rate responds to monetary policy too!

Increase in the Our Real Money Supply: **SR**

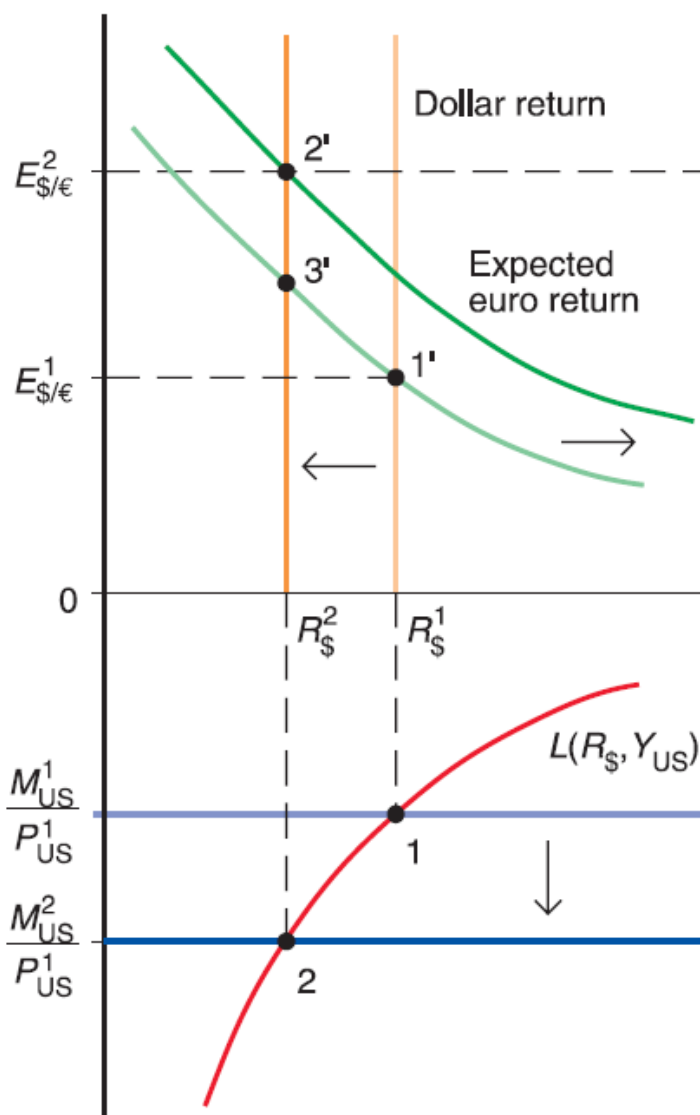


Increase in the Our Real Money Supply: LR



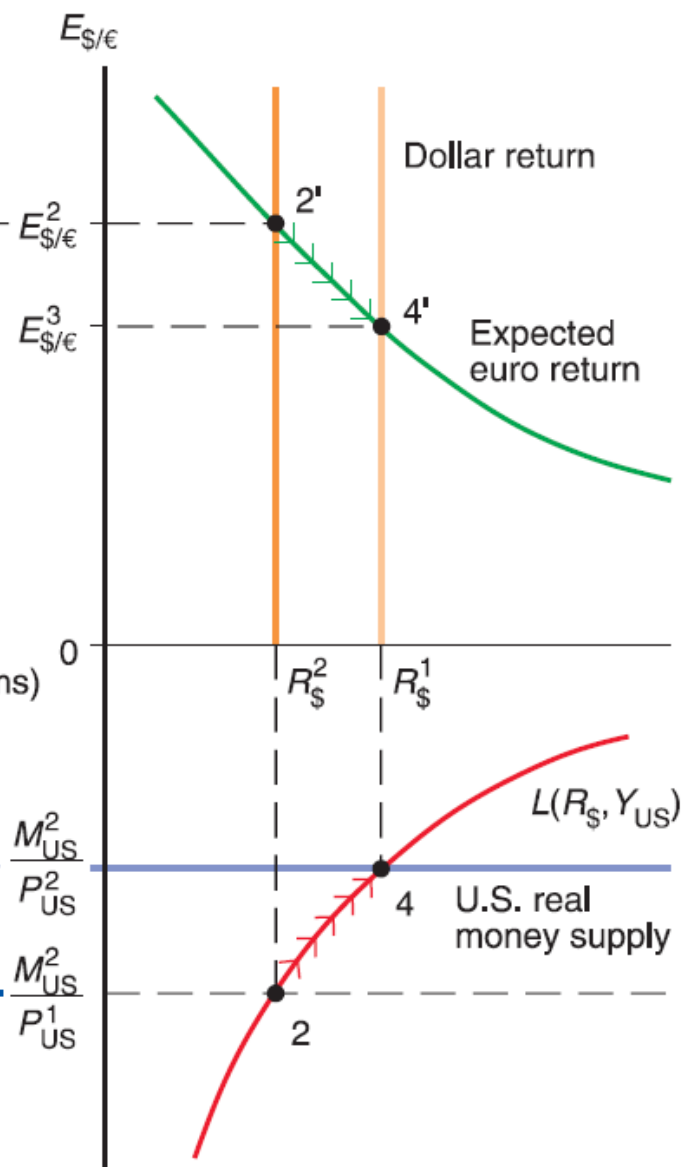
Money Supply and FOREX: SR & LR

Dollar/euro exchange rate, $E_{\$/\epsilon}$



U.S. real money holdings

SR

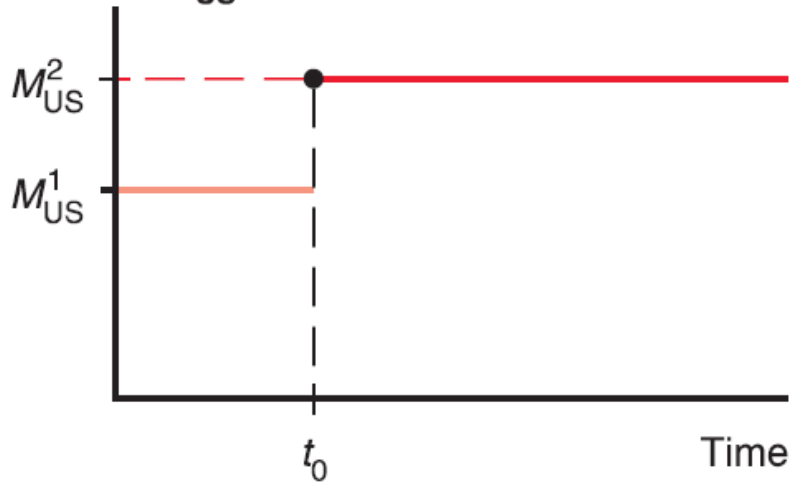


U.S. real money holdings

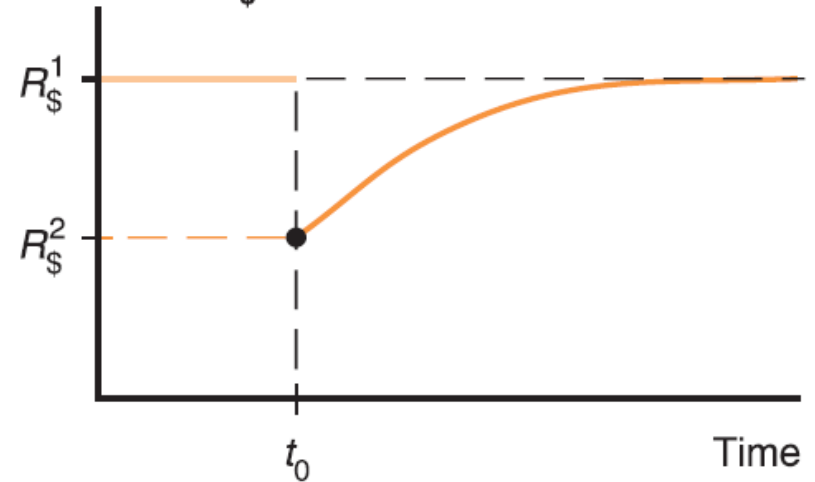
LR Adjustment

Effects of a Permanent Increase in the U.S. Money Supply: Dynamics

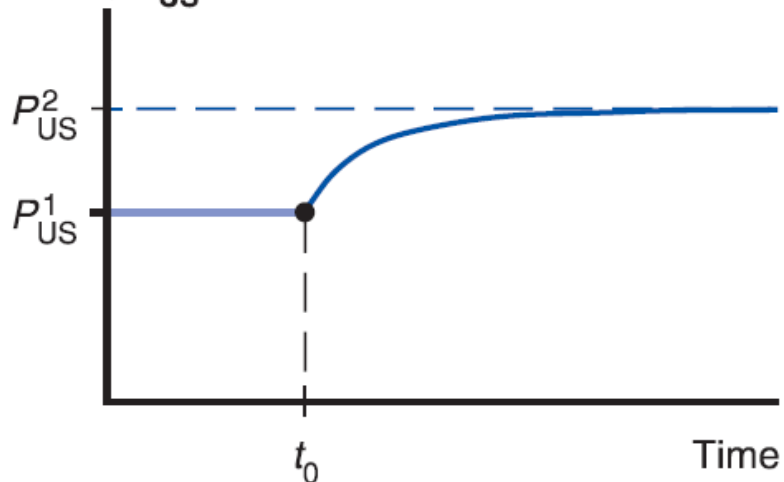
(a) U.S. money supply, M_{US}



(b) Dollar interest rate, $R_{\$}$



(c) U.S. price level, P_{US}

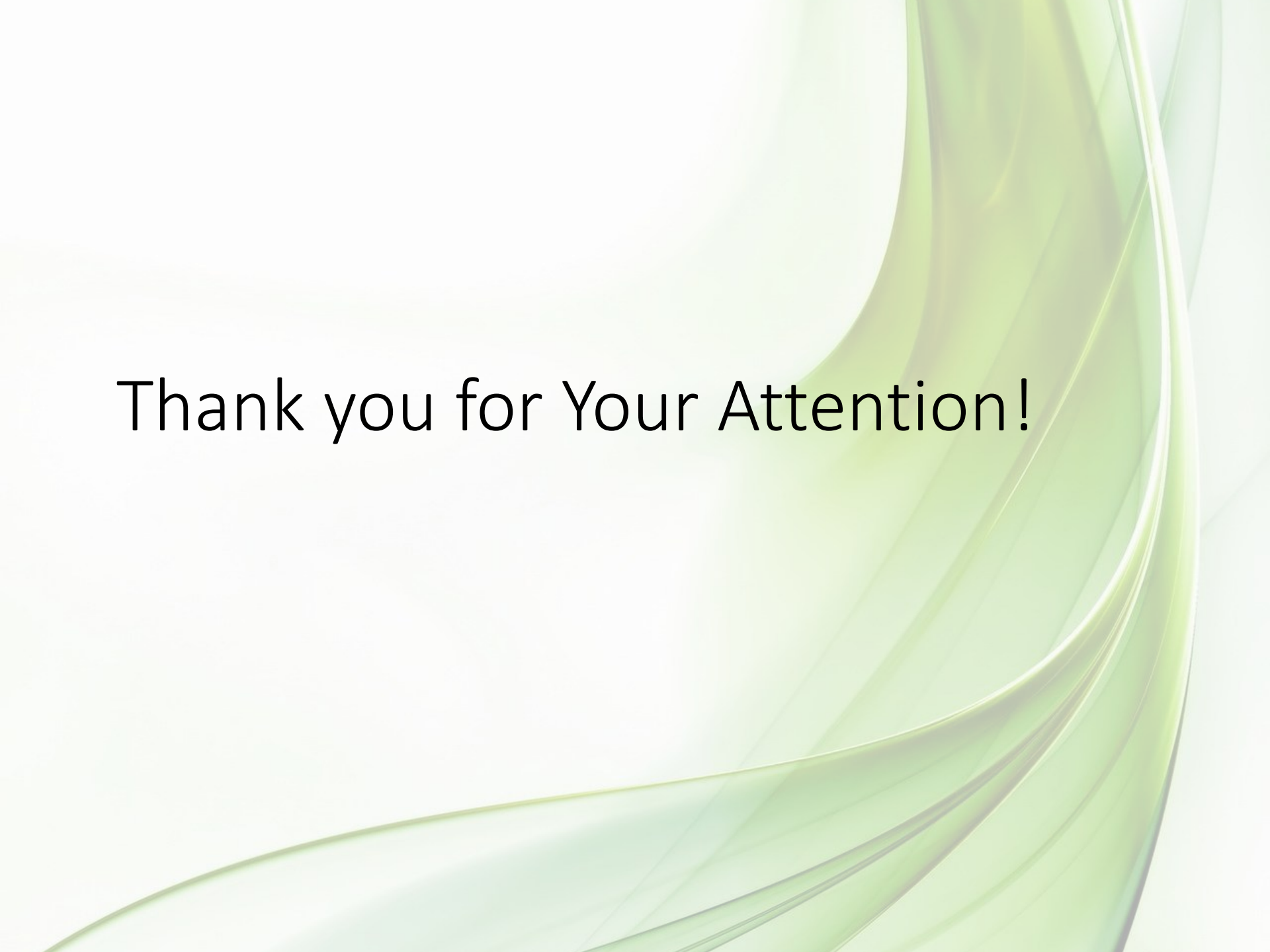


(d) Dollar/euro exchange rate, $E_{\$/\epsilon}$



Conclusions

- Short run effects are going to be more dynamic:
overshooting effect
 - If our central bank increases domestic money supply, exchange rate depreciates more than we neglected the effects on expectations
 - And after this initial sharp shock the exchange rate will gradually appreciate: LR effect will be more moderate
 - The subsequent appreciation is in fact necessary: interest rate will be converging to its equilibrium level (it will be gradually increasing)
 - Lower interest rate means the need for appreciation (UIP)

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Thank you for Your Attention!

References

- KOM – chapters 14, 15
- Agmon & Amihud (1981): The forward exchange rate and the prediction of the future spot rate: Empirical evidence. *Journal of Banking & Finance*, 1981, vol. 5, issue 3, 425-437
- Ha & Reddell (1998): What do forward interest and exchange rates tell us?